Weekly Planner

Week #10: (02/11/2026 To 06/11/2026)

Subject: Pre-Calculus Math **Unit Title: Functions, Graphs, Transformations, and Inverses** **Grade: 11**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1 Composite Functions	- I can evaluate and simplify composite functions. - I can determine the domain of a composite function.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 4.2 Ensure access to tools and assistive technologies.	- Khan Academy: Composite Functions - Desmos Graphing Calculator - Desmos Activity Builder
Lesson 2 Identifying One-to-One Functions	- I can determine whether a function is one-to-one using graphs, tables, and equations. - I can apply the Horizontal Line Test.		
Lesson 3 Finding the Inverse of a Function	- I can find the inverse of a function algebraically. - I can verify inverse functions using composition.		
Lesson 4 Graphing Inverse Functions	- I can graph a function and its inverse. - I can explain the relationship between a function and its inverse using symmetry about the line $y = x$.		
Homework	- Textbook Sections on Composite and Inverse Functions. - Complete assigned Desmos investigations. - Khan Academy practice on composition, one-to-one functions, and inverses.		
Teacher's notes/ Reminders	• Check **Microsoft Teams regularly** for class posts, announcements, resources, and assignments. • Complete assigned **Khan Academy** practice and extension activities. • Bring a charged device for graphing and technology-based investigations. • Review vocabulary including function, relation, continuity, limit, extrema, and rate of change. • Practice explaining solutions using appropriate mathematical notation and justification.		

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Technology Integration	• **Desmos.com**, Microsoft Teams, Graphing Calculator, GeoGebra
Cross-curricular links	**Computer Science:** Function composition resembles the use of multiple algorithms and programming functions. **Cybersecurity:** Encryption and decryption provide authentic examples of inverse processes. **Economics:** Input-output business models can be represented using composite functions. **Physics:** Reversing a process helps illustrate inverse relationships and mathematical modeling.
External Assessment Integration	Students will engage with: • **SAT/ACT/AP Precalculus-style questions involving function notation, composition, one-to-one functions, inverse functions, graphical analysis, and function modeling.** • **Multi-step problems requiring algebraic and graphical reasoning.**
Wellbeing/ ATL Skills	• **Critical Thinking:** Analyzing relationships between functions and their inverses. • **Communication Skills:** Explaining and justifying mathematical conclusions. • **Research Skills:** Using technology to investigate patterns and verify solutions. • **Self-Management:** Monitoring progress and selecting appropriate solving strategies. • **Thinking Skills:** Making connections among algebraic, graphical, numerical, and verbal representations.
Standards	• **CCSS.HSF-BF.A.1 Build functions from existing functions and model relationships.** • **CCSS.HSF-BF.B.4a Find inverse functions.** • **CCSS.HSF-BF.B.4b Verify inverses using composition.** • **CCSS.HSF-IF.C.7 Graph functions and identify key features.** • **CCSS.HSF-IF.B.4 Interpret key features of functions represented graphically and algebraically.** • **CCSS.HSF-LE.A.1 Analyze relationships and rates of change in function models.**
Inquiry/ TOK Question	• How can reversing a mathematical process help us better understand relationships and solve real-world problems?
Success Criteria (Measurable)	Lesson 1: Transformations of Functions ■ All students can Evaluate composite functions such as $f(g(x))$ ● Most students should Determine the domain of a composite function. ▲ Some students could Model a real-world process using function composition and explain its meaning. Lesson 2: Combining Transformations ■ All students can identify whether a function is one-to-one. ● Most students should apply the Horizontal Line Test correctly. ▲ Some students could explain Explain why one-to-one functions are necessary for inverses to exist.

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	Lesson 3: Function Operations and Composition ■ All students can Find the inverse of a simple function algebraically. ● Most students should Verify inverse functions using composition. ▲ Some students could Find inverses of more complex functions and interpret their meaning. Lesson 4: Inverse Functions ■ All students can Graph a function and its inverse. ● Most students should Identify reflection symmetry across the line $y = x$. ▲ Some students could Explain how domain and range are interchanged between a function and its inverse.
National Identity Domain	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Heritage التراث
	Area: UAE Innovation and Future Readiness Area: UAE culture and heritage Area: UAE Economic Development **Area:** Responsible Citizenship and Innovation
	Implementation: • Students explore how multiple processes in smart-city technologies can be represented using composite functions. • Students investigate how inverse operations are used in digital security, encryption, and authentication systems supporting the UAE's digital transformation initiatives. • Students examine real-life business systems in the UAE where outputs from one process become inputs to another process. • Students discuss how technology, data analysis, and innovation contribute to achieving the UAE Centennial 2071 vision and a knowledge-based economy.